AWS CloudFormation Student Lab Guide

Topic: VPC, Subnet, Security Group, EC2, and S3 (JSON Templates)
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1) Lab Objectives

- Understand Infrastructure as Code (IaC) with CloudFormation.
- Deploy network resources (VPC + Subnets + Routes + SG).
- Launch an EC2 instance from template parameters.
- Create and secure an S3 bucket using policy best practices.
- Learn reusable template design with Parameters and Outputs.

2) Required Files

- 00-all-in-one.json (single stack, beginner)
- 01-vpc-network.json (modular network)
- 02-ec2-instance.json (modular compute)
- 03-s3-bucket.json (modular storage)

3) Prerequisites

- AWS account with CloudFormation permissions.
- An existing EC2 Key Pair in your target region.
- AWS CLI configured (optional, if using CLI deployment).
- Deploy in one region consistently for all stacks.

4) Option A: One-Command Beginner Deployment

CLI Command

```
  aws cloudformation deploy \
    --stack-name student-all-in-one \
    --template-file 00-all-in-one.json \
    --parameter-overrides EnvironmentName=dev KeyPairName=YOUR_KEYPAIR
InstanceType=t3.micro BucketNamePrefix=travel237-allinone \
    --capabilities CAPABILITY_NAMED_IAM
```

Expected Result

- A VPC + public subnet + route to internet.
- A security group allowing SSH (22) and HTTP (80).
- An EC2 instance with Apache installed.
- An encrypted, versioned S3 bucket.

5) Option B: Modular Deployment

Step 1: Network Stack

```
  aws cloudformation deploy \
    --stack-name student-network \
    --template-file 01-vpc-network.json \
    --parameter-overrides EnvironmentName=dev VpcCidr=10.0.0.0/16
PublicSubnetCidr=10.0.1.0/24 PrivateSubnetCidr=10.0.2.0/24 AllowedSshCidr=0.0.0.0/0
```

Step 2: EC2 Stack (imports network outputs)

```
  aws cloudformation deploy \
    --stack-name student-ec2 \
    --template-file 02-ec2-instance.json \
    --parameter-overrides EnvironmentName=dev EnvironmentExportPrefix=dev
KeyPairName=YOUR_KEYPAIR InstanceType=t3.micro
```

Step 3: S3 Stack

```
aws cloudformation deploy \
  --stack-name student-s3 \
  --template-file 03-s3-bucket.json \
  --parameter-overrides EnvironmentName=dev BucketNamePrefix=travel237-demo
```

6) Validation Checklist

- CloudFormation stack status = CREATE_COMPLETE.
- EC2 has a public IP and instance status checks pass.
- Opening EC2 Public IP in browser shows Apache test page.
- S3 bucket exists with versioning enabled.
- S3 public access block is fully enabled.

7) Common Troubleshooting

- S3 bucket name conflict: change BucketNamePrefix parameter.
- Key pair error: verify KeyPairName exists in selected region.
- No internet from EC2: verify subnet route table has 0.0.0.0/0 via IGW.
- SSH blocked: confirm AllowedSshCidr and your local public IP.

8) Reusability Concepts to Teach

- Parameters make templates environment-agnostic.
- Outputs expose values for other stacks.
- Fn::ImportValue allows stack-to-stack composition.
- Tagging strategy improves operations and cost tracking.

9) Cleanup (Avoid Charges)

Delete in reverse dependency order for modular stacks:

- Delete student-ec2
- Delete student-s3
- Delete student-network

For all-in-one mode, delete student-all-in-one only.

CLI Cleanup Example

```
aws cloudformation delete-stack --stack-name student-all-in-one
```

10) Practice Extensions

- Add a private EC2 subnet with NAT gateway.
- Add IAM role for EC2 with least privilege access to S3.
- Convert to nested stacks.
- Add CloudWatch alarms for EC2 CPU.

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